

UniCare – AI Powered Complaint & Issue Resolution Platform

A Synopsis

for

Minor Project

BACHELOR OF TECHNOLOGY in COMPUTER SCIENCE & ENGINEERING

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Introduction

Universities often face challenges in managing and resolving complaints raised by students regarding infrastructure, hostel facilities, internet connectivity, electricity issues, and academic concerns. In most institutions, complaints are handled manually through emails, forms, or direct communication with authorities, which leads to delays, lack of transparency, and inefficient tracking of issues.

The **UniCare – AI Powered Complaint & Issue Resolution System** is designed to provide a centralized web-based platform where students can easily register complaints and track their resolution status. The system uses intelligent text analysis to automatically categorize complaints and determine their priority level based on the complaint description.

The proposed system aims to reduce manual intervention in complaint classification, improve response time, and provide administrators with analytical insights through dashboards and reports. The system is built using modern web technologies and follows a modular architecture that separates the AI logic into a dedicated service.

Literature Review

Several complaint management systems have been developed to improve service delivery and issue tracking in organizations and institutions. Traditional systems mainly rely on manual categorization and prioritization of complaints, which can lead to delays and human errors.

Recent research in **Natural Language Processing (NLP)** and **machine learning** has shown that textual complaints can be automatically classified using algorithms such as TF-IDF, Naive Bayes, and Logistic Regression. These techniques convert textual data into numerical vectors and enable automated classification of issues based on learned patterns.

Studies have also shown that automated complaint classification systems can significantly improve efficiency and response time compared to manual approaches. The proposed UniCare system adopts similar NLP concepts to enhance complaint categorization and priority prediction.

Problem Definition

In many universities, complaint handling is still performed through manual processes such as email communication, paper forms, or informal reporting. These approaches lead to several problems:

- Lack of a centralized complaint management system
- Delayed response to urgent issues
- Difficulty in tracking complaint status
- Manual categorization of complaints which is time-consuming
- Lack of data analytics for administrators

Therefore, there is a need for an automated system that can efficiently manage complaints, categorize them automatically, and assist administrators in prioritizing issues for faster resolution.

Objectives

The main objectives of the proposed system are:

- To develop a centralized platform for registering and managing university complaints.
- To automatically classify complaints into appropriate categories.
- To assign priority levels (High, Medium, Low) based on complaint severity.
- To provide real-time complaint tracking for users.
- To provide administrators with analytics and insights for decision making.
- To improve transparency and efficiency in the complaint resolution process.

Methodology

The proposed UniCare system follows a modular and scalable architecture that integrates frontend, backend, database, and AI components to ensure efficient complaint management and intelligent automation.

- **Frontend Development**

The user interface is developed using React.js and Tailwind CSS, providing a clean, responsive, and user-friendly experience for both students and administrators. The design focuses on accessibility, ease of navigation, and real-time interaction.

- **Backend Development**

The backend is built using Node.js and Express.js, which handle RESTful API requests, complaint processing, authentication, and system logic. It acts as a bridge between the frontend, database, and AI services.

- **Database Management**

MongoDB is used as a NoSQL database to efficiently store and manage complaint records, user data, and system analytics. Mongoose is utilized for schema modeling and data validation.

- **Complaint Classification**

Initially, the system uses a rule-based NLP approach based on keyword matching to automatically classify complaints into categories and assign priority levels. This ensures quick and lightweight processing.

- **AI Integration (Future Enhancement)**

To improve accuracy and scalability, the system is designed to integrate a machine learning model using TF-IDF vectorization and Logistic Regression. This model will be deployed via a FastAPI microservice and connected through REST APIs.

- **Analytics and Reporting**

The system includes an analytics dashboard for administrators, providing meaningful insights such as complaint distribution, resolution rates, and performance trends using data visualization techniques.

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